



LE QUANG SANG

Software Engineer (Full Stack, Systems & Applied AI)

Architecting resilient distributed systems, modern web platforms, mobile solutions & production-grade Applied AI

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Kim Bang, Ninh Binh, Vietnam

4+

Years of Experience

Software Engineering, Web & AI

10+

Production Systems

ERP, 3D Vision, CRM, FinTech

3.61

Graduation GPA

Summa Cum Laude / High Distinction

100%

Engineering Standard

Clean Code, Security & Performance

ENGINEERING SKILLS & COMPETENCIES

Software Engineering & Architecture

System Architecture

Clean Code & Design Patterns

RESTful APIs

OOP & Data Structures

High Concurrency & Caching

Database Optimization & Indexing

Application Security (OWASP)

Microservices Architecture

Applied AI & Computer Vision

YOLOv11

PyTorch

OpenCV

Open3D

TensorFlow

Keras

Scikit-Learn

Tesseract OCR

LLMs (Llama, DeepSeek)

AI Agents (Hermes, n8n)

Web Development (Full Stack)

JavaScript / TypeScript

Vue.js (Vue 3)

ReactJS

Next.js

PHP (Laravel)

Python (Django, Flask)

HTML5 / CSS3

Tailwind CSS

Bootstrap

Mobile App Development

Flutter (Dart)

React Native

Cross-platform Architecture

Offline-First Sync

Databases & In-Memory Stores

PostgreSQL

MySQL

MSSQL

SQLite

MongoDB

Firebase (Firestore/RTDB)

Redis

DevOps, Cloud & API Integrations

Docker

AWS (EC2, S3, Lambda, DynamoDB, Cognito, Amplify)

Google Cloud Platform

Cloudflare

Git / GitHub Actions

Meta Ads API

Google Ads API

Salesforce API

Languages & Engineering Mindset

Software Engineering Profile & Summary

Versatile Software Engineer graduated with Highest Distinction (Summa Cum Laude, GPA 3.61/4.0) from Electric Power University, combining strong theoretical computer science fundamentals with broad, real-world engineering execution. Experienced in architecting scalable, resilient systems: from complex distributed enterprise ERP platforms and open-source financial SaaS to cross-platform mobile apps and production-grade Computer Vision systems (3D pallet volume prediction with YOLOv11 & Open3D, biometric facial recognition). Pragmatic software engineer driven by high performance, application security (AppSec), clean maintainable code, and high-impact business outcomes.

Engineering Experience & Systems

All Projects

Systems & Full Stack

AI & Computer Vision

Mobile App

Zotek8

08/2023 - Present • Hanoi, Vietnam

Software Engineer (Full Stack & AI)

Lead software engineer responsible for system architecture, database design, and end-to-end delivery of complex enterprise applications, mobile clients, and applied AI systems.

Meeting Transcription Nagase (AI Speech-to-Text & Transcription)

Mobile, Cloud & AI Systems

Enterprise mobile audio recording and AI Speech-to-Text transcription suite with automated meeting summary generation and bidirectional sync with Salesforce CRM.

Key Responsibilities & Contributions:

- ✓ Built Flutter app: developed recording UI, audio streaming, and Offline-First SQLite caching.
- ✓ Engineered Serverless backend with AWS Lambda for audio processing and Salesforce CRM API integration.
- ✓ Configured user authentication, security, and access control policies via AWS Cognito.
- ✓ Deployed and optimized OpenAI Whisper on AWS EC2 for low-latency automated speech transcription.

- Automated end-to-end meeting minutes generation with secure bidirectional synchronization to Salesforce CRM REST APIs.
- Implemented an Offline-First architectural pattern with SQLite local queuing and automatic reconciliation.

Flutter (Dart)

AWS Lambda

AWS EC2

AWS Cognito

OpenAI Whisper

AWS Amplify

Salesforce API

SQLite

Camera-count (3D Multi-Camera Pallet AI)

AI & Computer Vision

Multi-angle 3D computer vision inspection system combining 3 synchronized cameras with geometric reconstruction to estimate volume and count occluded pallets in warehouses.

Team Leadership & Technical Planning (Led a 5-member engineering team)

Vietnamese (Native)

English (B1 - Technical Documentation & Communication)

Critical Thinking

Complex Problem Solving

Collaborative Teamwork

EDUCATION & CREDENTIALS

Electric Power University (EPU)

The Degree of Engineer in Information Technology

Major: Artificial Intelligence & Computer Vision

08/2019 - 04/2024

Honors: HIGH DISTINCTION (Xuất Sắc)

GPA: Cumulative 3.61 / 4.0

Graduated in the Top 10 of the department with strong mastery in algorithms, computer architecture, networking, and computer vision systems.

VIEW VERIFIED CREDENTIAL

The Degree of Engi...
Electric Power Univers...

Academic Transcri...
Courses 1 - 45: Core C...

Academic Transcri...
Courses 46 - 57: Thesi...

HONORS & ACHIEVEMENTS

3rd Prize (Team) - Innovation Challenge 2022

Faculty of IT - Electric Power University • 06/2022

Recognized for the project 'Biometric AI Face-Recognition Attendance & HR Management System'.

High Distinction Graduate (GPA 3.61/4.0)

Electric Power University • 04/2024

Honored among top-tier academic graduates in the Department of Information Technology.

PERSONAL INTERESTS

Distributed systems architecture, open-source software & AI agents

Running, cultivating endurance, grit & discipline

Strategic thinking & logic games

Key Responsibilities & Contributions:

✓ Researched, trained, and benchmarked YOLOv11 models combined with computer vision algorithms for precise pallet/package counting.

✓ Engineered multi-angle 3D geometric point cloud reconstruction with Open3D & OpenCV to resolve optical blind spots and stacking occlusions.

✓ Built an interactive 3D Base Demo (PoC) application using Streamlit & Plotly for real-time visualization and customer validation.

• Successfully eliminated optical blind spots and heavy stacking occlusions in live warehouse logistics.

• Delivered a production-ready PoC pipeline proving high feasibility before warehouse integration.

Python

YOLOv11

OpenCV

Open3D

PyTorch

StreamLit

Plotly

ERP Procurement System (Manufacturing Sector)

Enterprise Architecture

Enterprise procurement digitalization platform covering Purchase Requests (PR), Request for Quotation (RFQ), Purchase Orders (PO), inventory inspection, and financial settlements.

Key Responsibilities & Contributions:

✓ Collaborated with the Technical Leader to analyze client procurement business workflows and design the normalized MSSQL database schema.

✓ Scaffolded and developed the entire Frontend architecture from base using Vue 3, Vite & Tailwind CSS, implementing modular state management and multi-tier approval flows.

✓ Integrated RESTful APIs with Django DRF backend, optimizing high-volume data grid rendering and complex order form interactions.

• Successfully digitized the end-to-end manufacturing procurement lifecycle from PR, RFQ, PO to quality inspection and accounts payable.

• Leveraged Celery & Redis asynchronous task workers for heavy document batch exports, ensuring sub-second UI responsiveness.

Vue 3 (Vite)

Django DRF

MSSQL

Redis

CeLery

Docker

Tailwind CSS

InvoiceShelf (Open-Source FinTech SaaS)

FinTech & Open Source

Open-source invoice and cash flow management platform: automated billing, overdue tracking, high-speed PDF generation, EMVCo dynamic QR payments, and automated database backups.

Key Responsibilities & Contributions:

✓ Maintained system reliability, refactored core codebase, and resolved production bugs to ensure continuous uptime.

✓ Delivered end-to-end full-stack features: designed MySQL relational schemas, implemented Laravel REST APIs, and crafted frontend UI components.

✓ Optimized batch PDF invoice generation with DomPDF and automated recurring billing notifications via Laravel Queues & Scheduler.

• Automated recurring billing cycles and overdue alerts via queued background jobs.

• Integrated high-speed PDF rendering and dynamic banking QR codes complying with the EMVCo standard.

Laravel (PHP)

Vite

Tailwind CSS

MySQL

REST API

DomPDF

Laravel Queue

Docker

Adstart (Cross-Channel Ad Operations Hub)

Marketing Tech

Centralized ad campaign operations hub integrating Meta & Google Ads APIs: URL ROI attribution, real-time blacklisted keyword filtering, and automated financial reconciliations.

Key Responsibilities & Contributions:

✓ Maintained and monitored ad management infrastructure, ensuring stable 24/7 continuous campaign telemetry synchronization.

✓ Engineered end-to-end full-stack capabilities: designed MySQL schemas, built Laravel REST endpoints, and developed interactive reporting dashboards.

✓ Enhanced bidirectional Meta Marketing & Google Ads API integrations, upgrading policy-violating keyword filters and financial reconciliation queries.

• Deep bidirectional integration with Facebook Marketing API & Google Ads API for synchronized campaign metrics.

• Automated policy-violating keyword filtering engine protecting customer accounts from suspensions.

Laravel

Facebook Marketing API

Google Ads API

MySQL

Redis

Docker

Bootstrap

Kurua (Orthopedic Rehabilitation Healthcare Platform)

Healthcare & Mobile App

Digital healthcare platform tracking orthopedic rehabilitation regimens for specialty clinics, connecting physicians and patients seamlessly via Web and Mobile Apps.

Key Responsibilities & Contributions:

- ✓ Maintained system reliability, resolved production issues, and ensured continuous electronic health record data integrity.
 - ✓ Spearheaded end-to-end full-stack feature development: modeled MySQL relational databases and built Laravel REST APIs.
 - ✓ Implemented synchronized feature rollouts across the clinician web portal (Vue.js) and patient mobile application (React Native).
- Real-time rehabilitation progress and exercise synchronization between Doctor Web App and Patient Mobile App via Firebase.
 - Strict healthcare privacy access controls and data security compliance for sensitive medical records.

Vue.js

React Native

Laravel

MySQL

Firebase

REST API

Docker

Will Watch (Wildlife Geospatial Mapping & AI Platform)

GIS, Geo-AI & Community

Crowdsourced wildlife geospatial mapping platform combining real-time community location telemetry with AI species classification models.

Key Responsibilities & Contributions:

- ✓ Analyzed system requirements and designed real-time digital mapping system architecture.
 - ✓ Architected database schemas optimized for geospatial coordinates, location telemetry, and wildlife metadata.
 - ✓ Delivered full-stack implementation: developed backend business logic, the web portal (Next.js), and cross-platform mobile app (Flutter).
- Engineered real-time digital mapping workflows processing crowdsourced geospatial telemetry data.
 - Ensured seamless cross-platform synchronization between the Flutter mobile app and Next.js web portal.

Next.js

Flutter

Firebase

Torus Engineering

05/2022 - 05/2022

• Hanoi, Vietnam

Front End & Tooling Engineer

Developed high-precision interactive data labeling tools for computer vision research and packaged modular embeddable AI widgets for enterprise B2B clients.

Skin Annotation Tool

AI Data Tooling

Interactive polygonal annotation and scoring web application for dermatological analysis, producing standardized ground truth datasets for ML training pipelines.

Key Responsibilities & Contributions:

- ✓ Maintained the internal annotation tool, fixed UI bugs, and ensured smooth continuous labeling workflows for dermatology datasets.
 - ✓ Engineered new interactive Canvas features: polygonal lesion tracing, multi-point measurement scoring, and high-resolution zoom/pan.
 - ✓ Built Python Flask backend modules to validate and serialize annotation datasets into standardized JSON for ML training pipelines.
- Built high-performance, low-latency Canvas manipulation and geometric measurement tools.
 - Engineered standardized JSON dataset serialization for seamless machine learning pipelines.

Python Flask

JavaScript (ES6+)

Bootstrap

Canvas API

Belle Widget (Modular AI Component)

B2B Embeddable SDK

Packaged proprietary AI models into standalone embeddable Web/Mobile UI widgets for plug-and-play B2B client website integration.

Key Responsibilities & Contributions:

- ✓ Maintained embeddable AI widgets, ensuring high runtime stability, cross-browser compatibility, and zero conflicts with host environments.
 - ✓ Developed new features for Web (React) and Mobile (React Native) widget modules, supporting dynamic thematic customization for B2B clients.
 - ✓ Integrated and synchronized API client communication with backend services (.NET and Python) to render real-time AI inference results.
- Architected isolated component sandboxing preventing CSS/JS namespace collisions when injected into third-party sites.

React React Native JavaScript Python .NET API

R&D Lab – Electric Power University
Software R&D Engineer

2020 - 2024 • Hanoi, Vietnam

Key contributor to foundational computer science research initiatives, building applied deep learning and real-time computer vision systems.

Biometric Face Recognition Attendance & HR System

AI & Award Winning

Comprehensive enterprise attendance and identity verification platform powered by deep learning facial feature extraction and real-time biometric matching.

- Awarded 3RD PRIZE in the prestigious Innovation Challenge 2022 by EPU Faculty of IT.
- Achieved sub-500ms facial verification latency with enterprise-grade accuracy.

AnguLar .NET Core Python C++ SQL Server Face Recognition

Document OCR & Intelligent Image Retrieval

OCR & Document Processing

Optical Character Recognition web app converting scanned documents and PDFs into searchable, indexable full text.

- Developed adaptive image pre-processing (binarization, skew correction) drastically boosting recognition accuracy.

Python Flask Tesseract OCR 5 OpenCV JavaScript

Independent Products & Open Source (Personal Projects)
Creator & Lead Software Engineer

2024 - Present • Hanoi, Vietnam / Remote

Architecting, developing, and operating independent production-grade software applications, demonstrating deep engineering capabilities across full-stack architecture, privacy-first client-side systems, cutting-edge AI SEO, and edge AI.

OmniKit Web Tool (54+ Client-Side Online Utility Suite)

Client-Side Suite / AI SEO

omnikit.online GitHub

Comprehensive suite of 53 free online utility tools (Typing speed test, text diff checker, developer utilities for JSON/JWT/Regex/Hash, image/PDF processing, financial ROI calculators, WCAG color contrast, QR generator). 100% client-side execution ensuring zero user data ever touches external servers.

- Engineered with cutting-edge AI SEO & GEO First architecture: Answer-First content modeling, native /llms.txt for AI crawlers, schema-dts structured data, and static generation (SSG) across 341 pages.
- Integrated Firebase Auth for unlimited usage tiers and full internationalization across 5 languages (VI, EN, ZH, JA, KO).

Next.js (SSG) TypeScript Tailwind CSS Firebase Auth Web Crypto API
Web Workers Schema.org (JSON-LD)

SangLe AI Showcase (Real-Time In-Browser AI Platform)

Real-Time Browser AI

sangle-ai.vercel.app GitHub

High-performance real-time browser AI showcase platform powered by Google MediaPipe Tasks, leveraging WebAssembly (WASM) and WebGL hardware acceleration across GPU/CPU for ultra-low latency on-device intelligence.

- Deployed 15+ frontier AI models across 3 core modalities: Computer Vision (468+ point Face Landmarker with blendshapes, Hand tracking, Gesture recognition, Pose estimation, 1-click Interactive Segmentation), NLP (client-side LLM inference with Gemma), and Audio classification.
- 100% On-Device edge inference with zero camera/mic streaming to the cloud, ensuring complete privacy and high responsiveness on desktop and mobile.

Next.js 16 (App Router) TypeScript Google MediaPipe WASM WebGL
Tailwind CSS Radix UI Chart.js